NAME : SUBASH CHANDRA BOSE

DATE : 23.02.2026

Integration of Snort IDS with Wazuh SIEM for Real-Time Intrusion Detection and Monitoring

1. Aim

To configure and integrate Snort Intrusion Detection System (IDS) with Wazuh SIEM to monitor and analyze network traffic between virtual machines and detect malicious activities such as ICMP ping and HTTP access attempts.

2. Objectives

- Install and configure Snort 3 on Kali Linux.

- Create custom detection rules for ICMP and HTTP traffic.

- Generate network traffic from Ubuntu machine.

- Configure Wazuh agent to ingest Snort logs.

- Visualize IDS alerts in Wazuh Dashboard.

- Simulate a basic Security Operations Center (SOC) environment.

3. Lab Environment

Attacker VM: Ubuntu Linux

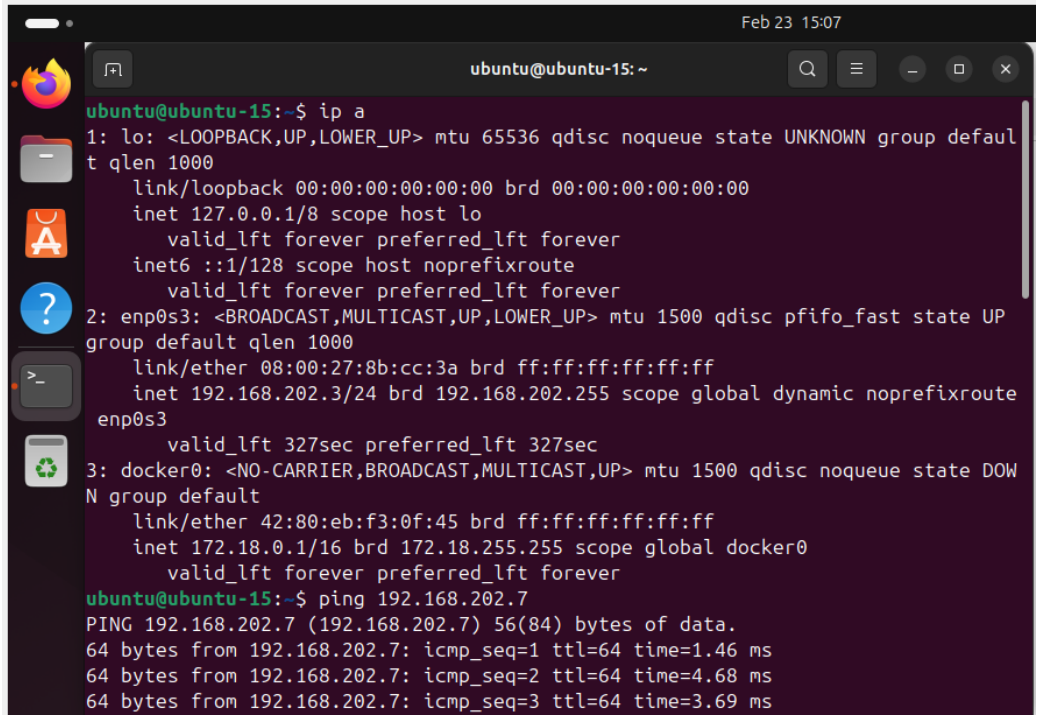
IDS VM: Kali Linux

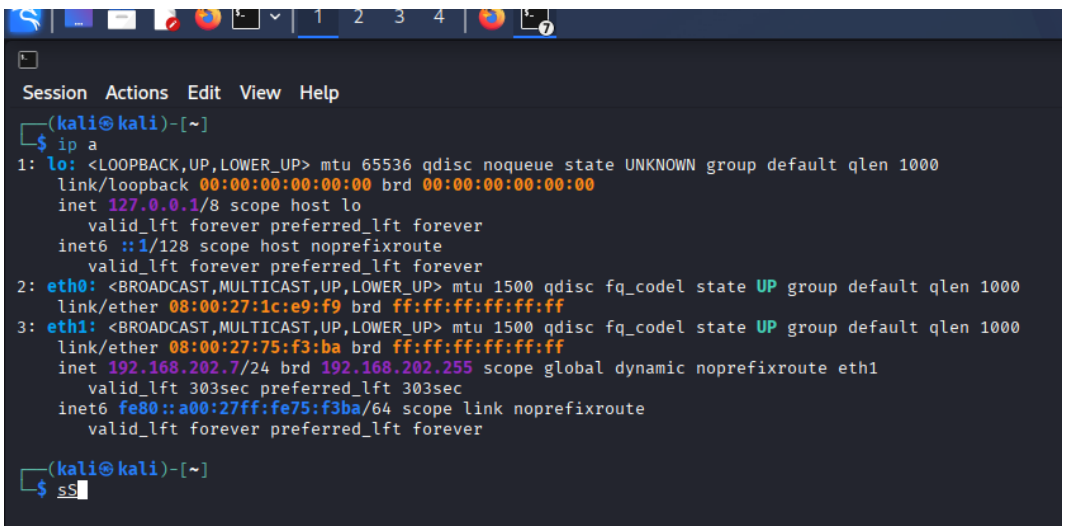
SIEM: Wazuh Manager & Dashboard

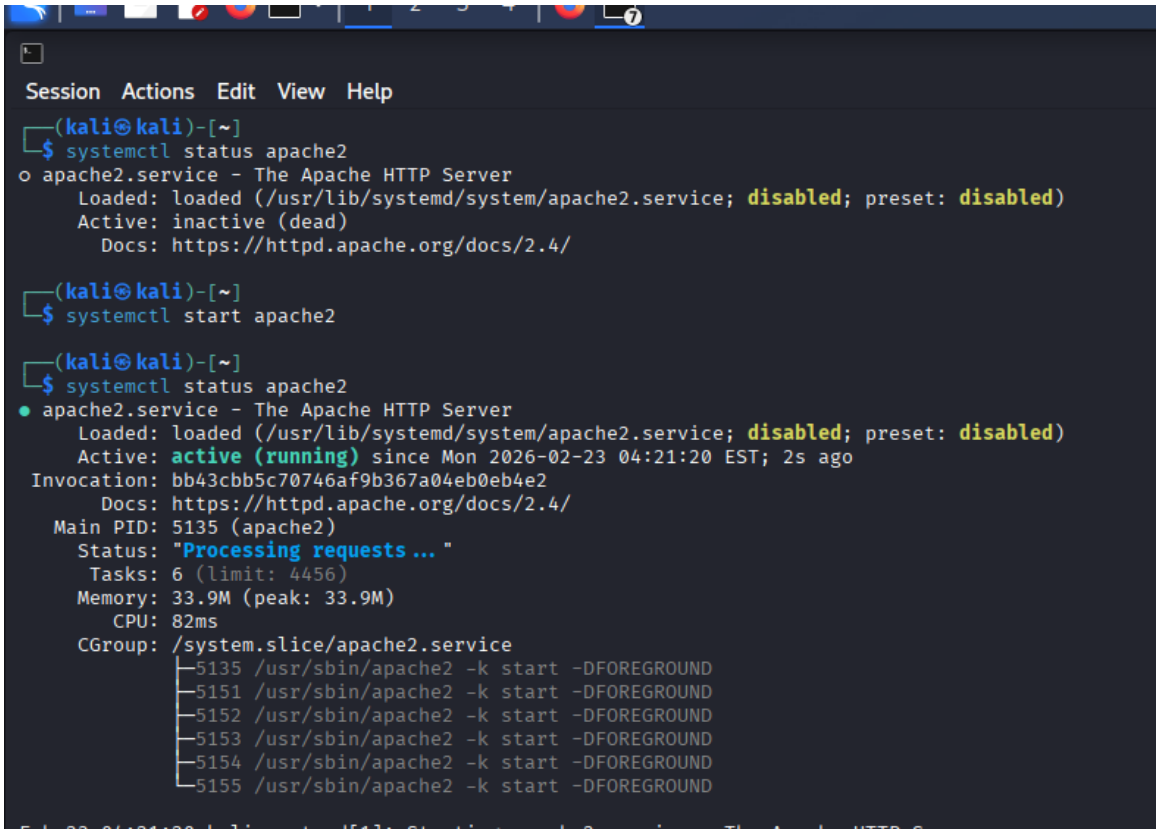
IDS Tool: Snort 3

Web Server: Apache hosted on Kali Linux

Network Mode: Host-Only (monitoring) and NAT (communication with Wazuh)


Feb 23 15:07
ubuntu@ubuntu-15: ~
ubuntu@ubuntu-15:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defaul
t qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP
group default qlen 1000
link/ether 08:00:27:8b:cc:3a brd ff:ff:ff:ff:ff:ff
inet 192.168.202.3/24 brd 192.168.202.255 scope global dynamic noprefixroute
enp0s3
valid_lft 327sec preferred_lft 327sec
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOW
N group default
link/ether 42:80:eb:f3:0f:45 brd ff:ff:ff:ff:ff:ff
inet 172.18.0.1/16 brd 172.18.255.255 scope global docker0
valid_lft forever preferred_lft forever
ubuntu@ubuntu-15:~$ ping 192.168.202.7
PING 192.168.202.7 (192.168.202.7) 56(84) bytes of data.
64 bytes from 192.168.202.7: icmp_seq=1 ttl=64 time=1.46 ms
64 bytes from 192.168.202.7: icmp_seq=2 ttl=64 time=4.68 ms
64 bytes from 192.168.202.7: icmp_seq=3 ttl=64 time=3.69 ms


Session Actions Edit View Help
(kali㉿kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:1c:e9:f9 brd ff:ff:ff:ff:ff:ff
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:75:f3:ba brd ff:ff:ff:ff:ff:ff
inet 192.168.202.7/24 brd 192.168.202.255 scope global dynamic noprefixroute eth1
valid_lft 303sec preferred_lft 303sec
inet6 fe80::a00:27ff:fe75:f3ba/64 scope link noprefixroute
valid_lft forever preferred_lft forever
(kali㉿kali)-[~]
$ ss



4. Architecture Overview

Ubuntu (Attacker) → Kali (Snort IDS + Apache + Wazuh Agent) → Wazuh Manager → Wazuh Dashboard

5. Procedure

Step 1: Network Configuration

Both Ubuntu and Kali virtual machines were configured in Host-Only mode to allow traffic monitoring. NAT mode was later used to allow Kali to communicate with the Wazuh Manager.

Step 2: Snort Installation and Verification

Snort 3 was installed on Kali Linux and verified using the command: snort -V. The main configuration file used was /etc/snort/snort.lua.

Step 3: Custom Snort Rules Creation

ICMP Detection Rule:

alert icmp any any -> any any (msg:"ICMP Ping Detected"; classtype:attempted-recon; sid:1000001; rev:3;)

HTTP Detection Rule:

alert tcp any any -> any 80 (msg:"HTTP Access Detected"; flow:to_server,established; content:"GET"; http_method; classtype:web-application-activity; sid:1000002; rev:1;)

```
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ sudo nano /etc/snort/snort.lua
[sudo] password for kali:

┌──(kali㉿kali)-[~]
└─$ sudo nano /etc/snort/local.rules

┌──(kali㉿kali)-[~]
└─$ cat /etc/snort/local.rules
alert icmp any any → any any ( msg:"ICMP Ping Detected"; sid:1000001; rev:1; )
alert tcp any any → any 80 ( msg:"HTTP Access Detected"; sid:1000002; rev:1; )

┌──(kali㉿kali)-[~]
└─$
```

```
--------------------------------------------------------------------------------
-- 5. configure detection
--------------------------------------------------------------------------------

references = default_references
classifications = default_classifications

ips =
{
    -- use this to enable decoder and inspector alerts
    --enable_builtin_rules = true,
    rules = [[
        include /etc/snort/local.rules
      ]],

    -- use include for rules files; be sure to set your path
    -- note that rules files can include other rules files
    -- (see also related path vars at the top of snort_defaults.lua)

    variables = default_variables
}

-- use these to configure additional rule actions
-- react = { }
-- reject = { }

-- use this to enable payload injection utility
-- payload_injector = { }

--------------------------------------------------------------------------------
-- 6. configure filters
--------------------------------------------------------------------------------

-- below are examples of filters
```

Step 4: Snort Logging Configuration

Snort was configured to log alerts in JSON format to /var/log/snort/alert_json.txt.

Snort was started using: sudo snort -c /etc/snort/snort.lua -i eth1

```
┌──(kali㉿kali)-[~]
└─$ sudo snort -c /etc/snort/snort.lua -i eth1 -A alert_fast
--------------------------------------------------
o")~   Snort++ 3.10.2.0
--------------------------------------------------
Loading /etc/snort/snort.lua:
Loading snort_defaults.lua:
Finished snort_defaults.lua:
        ips
        file_id
        http2_inspect
        ftp_data
        port_scan
        gtp_inspect
        dce_http_server
        dce_http_proxy
        dce_udp
        dce_tcp
        dce_smb
        s7commplus
        opcua
        modbus
        mms
        iec104
        dnp3
        cip
        telnet
        ssl
        ssh
        sip
        rpc_decode
        pop
        normalizer
        netflow
        imap
        dns
        back_orifice
        arp_spoof
        stream_file
        stream_user
        stream_udp
        stream_tcp
        stream_icmp
        stream_ip
        stream
        active
        alerts
```

Session Actions Edit View Help

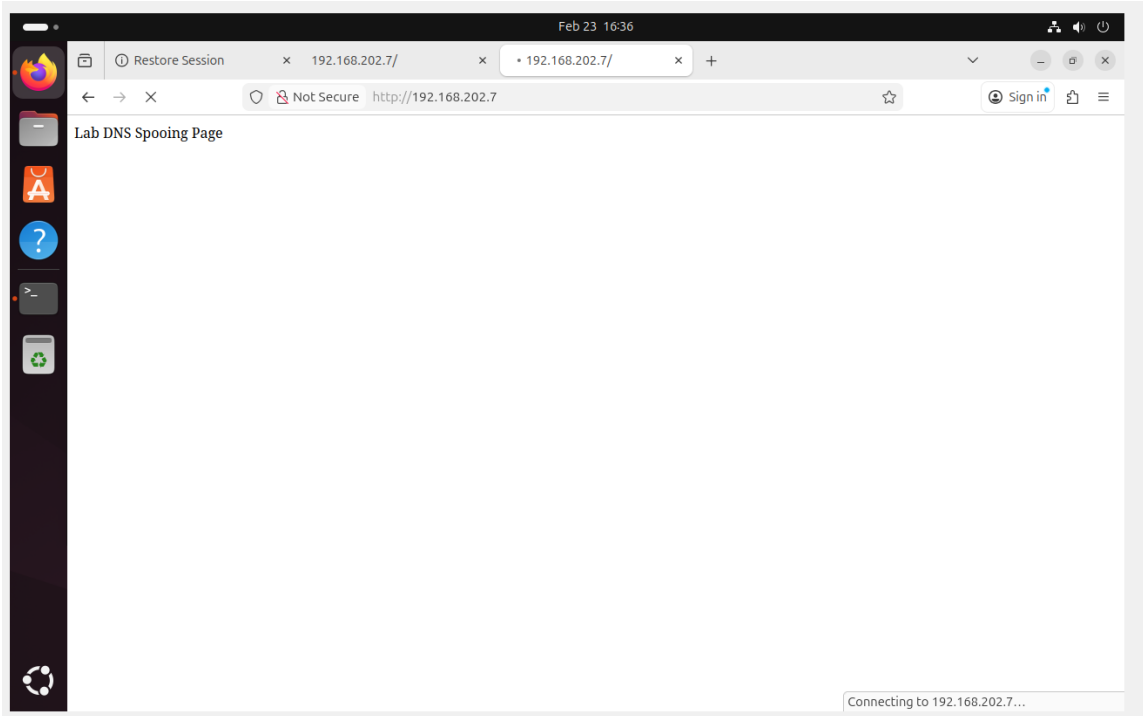
```
        output
Finished /etc/snort/snort.lua:
Loading file_id.rules_file:
Loading file_magic.rules:
Finished file_magic.rules:
Finished file_id.rules_file:
Loading ips.rules:
Loading /etc/snort/local.rules:
Finished /etc/snort/local.rules:
Finished ips.rules:
--------------------------------------------------
ips policies rule stats
               id  loaded  shared enabled    file
                0     221       0     221    /etc/snort/snort.lua
--------------------------------------------------
rule counts
       total rules loaded: 221
               text rules: 221
            option chains: 221
            chain headers: 3
--------------------------------------------------
port rule counts
             tcp     udp    icmp      ip
     any       0       0       1       0
     dst       1       0       0       0
   total       1       0       1       0
--------------------------------------------------
service rule counts          to-srv  to-cli
                  file_id:      219     219
                    total:      219     219
--------------------------------------------------
fast pattern groups
                to_server: 1
                to_client: 1
--------------------------------------------------
search engine (ac_bnfa)
                instances: 2
                 patterns: 438
            pattern chars: 2602
               num states: 1832
         num match states: 392
             memory scale: KB
             total memory: 71.2812
           pattern memory: 19.6484
        match list memory: 28.4375
        transition memory: 22.9453
appid: MaxRss diff: 2944
appid: patterns loaded: 300
--------------------------------------------------
pcap DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] eth1
```

```
kali@kali: ~
Session  Actions  Edit  View  Help
            total memory: 71.2812
          pattern memory: 19.6484
       match list memory: 28.4375
       transition memory: 22.9453
appid: MaxRss diff: 2944
appid: patterns loaded: 300

pcap DAQ configured to passive.
Commencing packet processing
Retry queue interval is: 200 ms
++ [0] eth1
02/23-04:33:13.093310 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:13.099866 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:13.099870 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:13.106542 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:13.800363 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:13.802357 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:13.983493 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:13.986606 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:14.154554 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:14.155879 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:14.549003 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:14.550281 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:14.748064 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:14.750086 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:14.920307 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:14.922721 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:15.084591 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:15.085647 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:15.289407 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:15.290622 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:20.300224 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:30.313008 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:47534 -> 192.168.202.7:80
02/23-04:33:30.313604 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:47534 -> 192.168.202.7:80
02/23-04:33:30.313605 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:47534 -> 192.168.202.7:80
02/23-04:33:30.315675 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:47534 -> 192.168.202.7:80
02/23-04:33:30.316115 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:47534 -> 192.168.202.7:80
02/23-04:33:30.316778 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:47534 -> 192.168.202.7:80
02/23-04:34:02.068962 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} fe80::3553:4885:9b7b:f2ae -> ff02::16
02/23-04:34:02.079163 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} fe80::3553:4885:9b7b:f2ae -> ff02::16
02/23-04:34:02.079203 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} fe80::3553:4885:9b7b:f2ae -> ff02::16
02/23-04:34:02.079320 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} fe80::3553:4885:9b7b:f2ae -> ff02::16
02/23-04:34:02.268898 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} fe80::3553:4885:9b7b:f2ae -> ff02::16
02/23-04:36:26.518115 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.3 -> 192.168.202.7
02/23-04:36:26.518312 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.7 -> 192.168.202.3
02/23-04:36:27.677955 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.3 -> 192.168.202.7
02/23-04:36:27.678037 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.7 -> 192.168.202.3
02/23-04:36:29.808353 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.3 -> 192.168.202.7
02/23-04:36:29.808429 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.7 -> 192.168.202.3
02/23-04:36:30.961920 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.3 -> 192.168.202.7
02/23-04:36:30.961995 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.7 -> 192.168.202.3
02/23-04:36:32.041841 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.3 -> 192.168.202.7
02/23-04:36:32.041938 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.7 -> 192.168.202.3
```

Step 5: Traffic Generation

ICMP test from Ubuntu: ping <Kali-IP>

HTTP test from Ubuntu: curl http://<Kali-IP>



Step 6: Verification of Snort Alerts

Alerts were successfully written to /var/log/snort/alert_json.txt and verified using the tail command.

```
++ [0] eth1
02/23-04:33:13.093310 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:13.099866 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:13.099870 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:13.106542 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:13.800363 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:13.802357 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:13.983493 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:13.986606 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:14.154554 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:14.155879 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:14.549003 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:14.550281 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:14.748064 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:14.750086 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:14.920307 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:14.922721 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:15.084591 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:15.085647 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:15.289407 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:15.290622 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:20.300224 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:57462 -> 192.168.202.7:80
02/23-04:33:30.313008 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:47534 -> 192.168.202.7:80
02/23-04:33:30.313604 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:47534 -> 192.168.202.7:80
02/23-04:33:30.313605 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:47534 -> 192.168.202.7:80
02/23-04:33:30.315675 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:47534 -> 192.168.202.7:80
02/23-04:33:30.316115 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:47534 -> 192.168.202.7:80
02/23-04:33:30.316778 [**] [1:1000002:1] "HTTP Access Detected" [**] [Priority: 0] {TCP} 192.168.202.3:47534 -> 192.168.202.7:80
02/23-04:34:02.068962 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} fe80::3553:4885:9b7b:f2ae -> ff02::16
02/23-04:34:02.079163 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} fe80::3553:4885:9b7b:f2ae -> ff02::16
02/23-04:34:02.079203 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} fe80::3553:4885:9b7b:f2ae -> ff02::16
02/23-04:34:02.079320 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} fe80::3553:4885:9b7b:f2ae -> ff02::16
02/23-04:34:02.268898 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} fe80::3553:4885:9b7b:f2ae -> ff02::16
02/23-04:36:26.518115 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.3 -> 192.168.202.7
02/23-04:36:26.518312 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.7 -> 192.168.202.3
02/23-04:36:27.677955 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.3 -> 192.168.202.7
02/23-04:36:27.678037 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.7 -> 192.168.202.3
02/23-04:36:29.808353 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.3 -> 192.168.202.7
02/23-04:36:29.808429 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.7 -> 192.168.202.3
02/23-04:36:30.961920 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.3 -> 192.168.202.7
02/23-04:36:30.961995 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.7 -> 192.168.202.3
02/23-04:36:32.041841 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.3 -> 192.168.202.7
02/23-04:36:32.041938 [**] [1:1000001:1] "ICMP Ping Detected" [**] [Priority: 0] {ICMP} 192.168.202.7 -> 192.168.202.3
```

Step 7: Wazuh Agent Configuration

The file /var/ossec/etc/ossec.conf was modified to include monitoring of /var/log/snort/alert_json.txt using JSON log format.

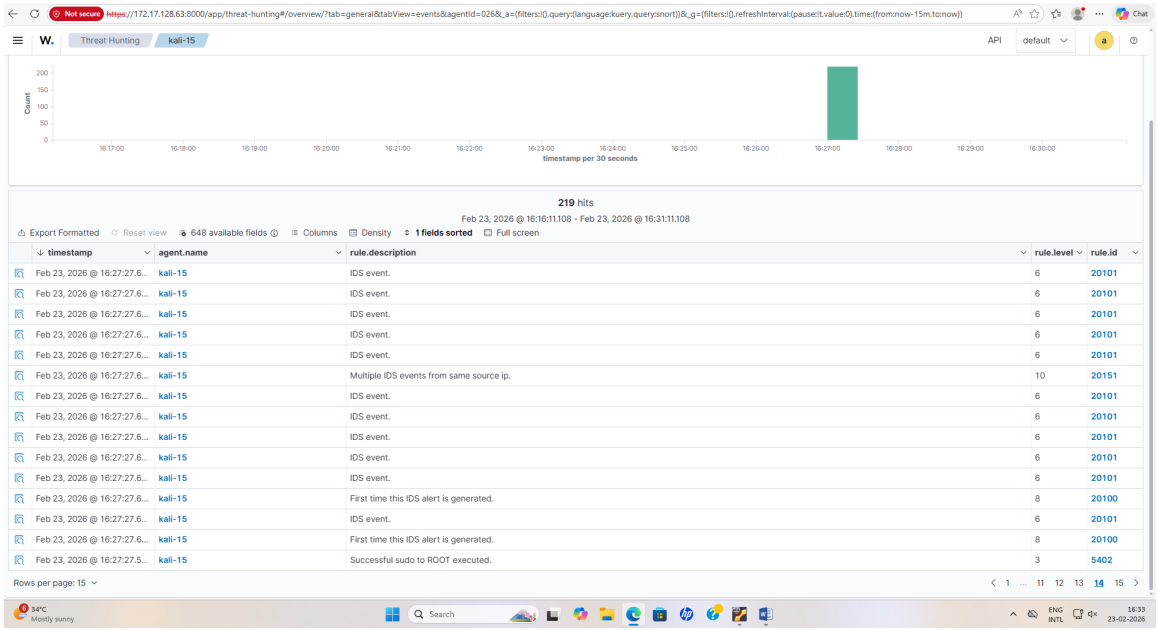
```
-- 7. configure outputs
------------------------------------------------------------------------------

-- event logging
-- you can enable with defaults from the command line with -A <alert_type>
-- uncomment below to set non-default configs
--alert_csv = { }
-- alert_fast = { }
--alert_full = { }
--alert_sfsocket = { }
alert_syslog =
 {
 }
--unified2 = { }
-- In snort.lua
alert_json = {
    file = true,
    limit = 10,
    fields = { 'timestamp', 'msg', 'proto', 'src_addr', 'src_port', 'dst_addr', 'dst_port', 'sid', 'rev', 'class' }
}

-- packet logging
-- you can enable with defaults from the command line with -L <log_type>
--log_codecs = { }
--log_hext = { }
--log_pcap = { }

-- additional logs
--packet_capture = { }
--file_log = { }
```

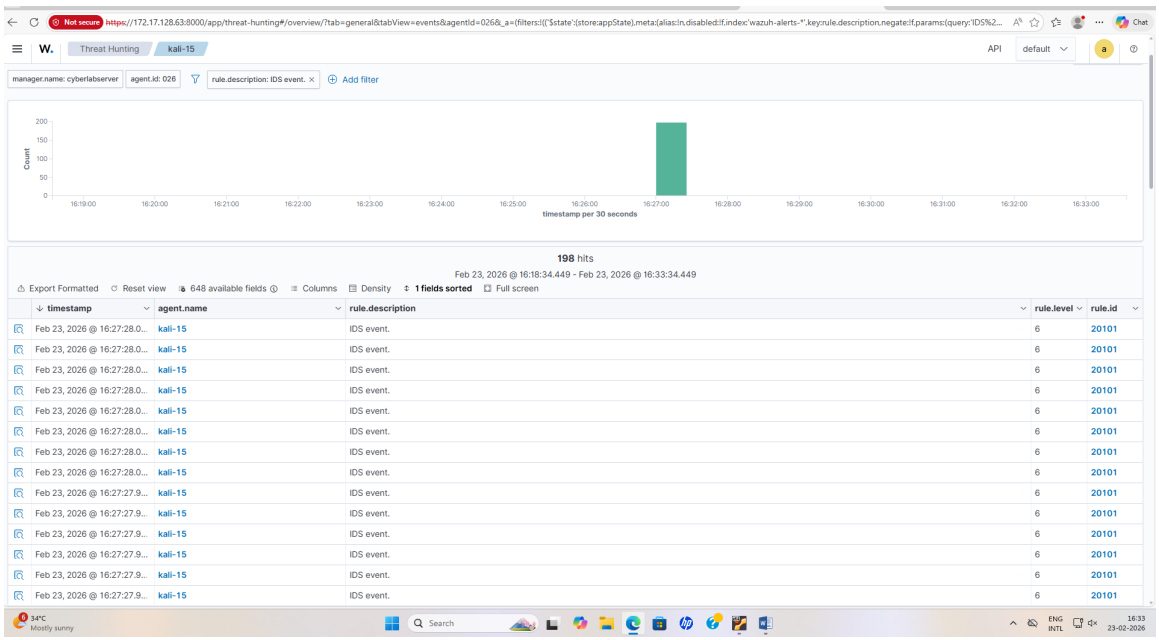
```
  <!-- Log analysis -->
  <localfile>
    <log_format>command</log_format>
    <command>df -P</command>
    <frequency>360</frequency>
  </localfile>
  <!-- snort -->
  <localfile>
     <log_format>snort-full</log_format>
     <location>/var/log/snort/alert_json.txt</location>
   </localfile>

  <localfile>
    <log_format>full_command</log_format>
    <command>netstat -tulpn | sed 's/\([[:alnum:]]\+\)\ \+[[:digit:]]
/:\1/' | sed 1,2d</command>
    <alias>netstat listening ports</alias>
    <frequency>360</frequency>
  </localfile>

  <localfile>
```

The Wazuh agent was restarted using: sudo systemctl restart wazuh-agent

Step 8: Viewing Alerts in Wazuh Dashboard

Alerts were viewed in the Wazuh Dashboard under Security Events → Discover. Filtering was performed using agent.name: kali-15.





6. Results

Snort successfully detected ICMP ping and HTTP access attempts. Alerts were logged in JSON format and ingested by Wazuh. The Wazuh dashboard successfully displayed IDS events from the Kali machine.

Document Details

View surrounding documents

View single document

Table JSON

_index	wazuh-alerts-4.x-2026.02.23
agent.id	026
agent.ip	10.0.2.15
agent.name	kali-15
data.dstip	192.168.202.3
data.id	1:1000001:1
data.srcip	192.168.202
decoder.name	snort
decoder.parent	snort
full_log	Feb 23 10:42:45 kali snort[9299]: [1:1000001:1] "ICMP Ping Detected" {ICMP} 192.168.202.7 -> 192.168.202.3
id	1771844248.14616619
input.type	log
location	journald
manager.name	cyberlabserver
predecoder.hostname	kali
predecoder.program_name	snort
predecoder.timestamp	Feb 23 10:42:45
rule.description	IDS event.
rule.firedtimes	81
rule.groups	ids
rule.id	20101
rule.level	6
rule.mail	false
timestamp	Feb 23, 2026 @ 16:27:28.089

Document Details

View surrounding documents

View single document

Table JSON

_index	wazuh-alerts-4.x-2026.02.23
agent.id	026
agent.ip	10.0.2.15
agent.name	kali-15
data.dstip	192.168.202.7:80
data.id	1:1000002:1
data.srcip	192.168.202.3
decoder.name	snort
decoder.parent	snort
full_log	Feb 23 10:41:51 kali snort[9299]: [1:1000002:1] "HTTP Access Detected" {TCP} 192.168.202.3:43928 -> 192.168.202.7:80
id	1771844247.14601578
input.type	log
location	journald
manager.name	cyberlabserver
predecoder.hostname	kali
predecoder.program_name	snort
predecoder.timestamp	Feb 23 10:41:51
rule.description	IDS event.
rule.firedtimes	37
rule.groups	ids
rule.id	20101
rule.level	6
rule.mail	false
timestamp	Feb 23, 2026 @ 16:27:27.705

Discover | wazuh-alerts-4.x-2026.02.23#EKYlipwBtP4lsdxOpszq

Table JSON

@timestamp	Feb 23, 2026 @ 16:27:27.705
_index	wazuh-alerts-4.x-2026.02.23
agent.id	026
agent.ip	10.0.2.15
agent.name	kali-15
data.dstip	192.168.202.7:80
data.id	1:1000002:1
data.srcip	192.168.202.3
decoder.name	snort
decoder.parent	snort
full_log	Feb 23 10:41:51 kali snort[9299]: [1:1000002:1] "HTTP Access Detected" {TCP} 192.168.202.3:43928 -> 192.168.202.7:80
id	1771844247.14601578
input.type	log
location	journald
manager.name	cyberlabserver
predecoder.hostname	kali
predecoder.program_name	snort
predecoder.timestamp	Feb 23 10:41:51
rule.description	IDS event.
rule.firedtimes	37
rule.groups	ids
rule.id	20101
rule.level	6
rule.mail	false
timestamp	Feb 23, 2026 @ 16:27:27.705

7. Conclusion

The integration of Snort IDS with Wazuh SIEM was successfully implemented. Custom rules were created to detect ICMP and HTTP traffic. Alerts were generated, logged, ingested, and visualized in the Wazuh dashboard, simulating a real-world SOC monitoring workflow.

8. Learning Outcomes

- Practical understanding of IDS configuration.

- Experience in SIEM log ingestion and visualization.

- Implementation of rule-based intrusion detection.

- Hands-on experience in SOC lab setup.